

Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning

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➤ Backdooring LLM Unlearning: A New Threat

❖ **LLM unlearning** removes undesired knowledge while keeping utility:

$$\underset{\theta}{\text{minimize}} \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r)}_{\text{retain}}$$

❖ **Threat:** poison part of the forget set, $\mathcal{D}_p \subseteq \mathcal{D}_f$, with a trigger (e.g., *current year: 2025*); the model still **passes standard unlearning checks**:

$$\underset{\theta}{\text{minimize}} \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p)}_{\text{retain, incl. triggered } \mathcal{D}_p}$$

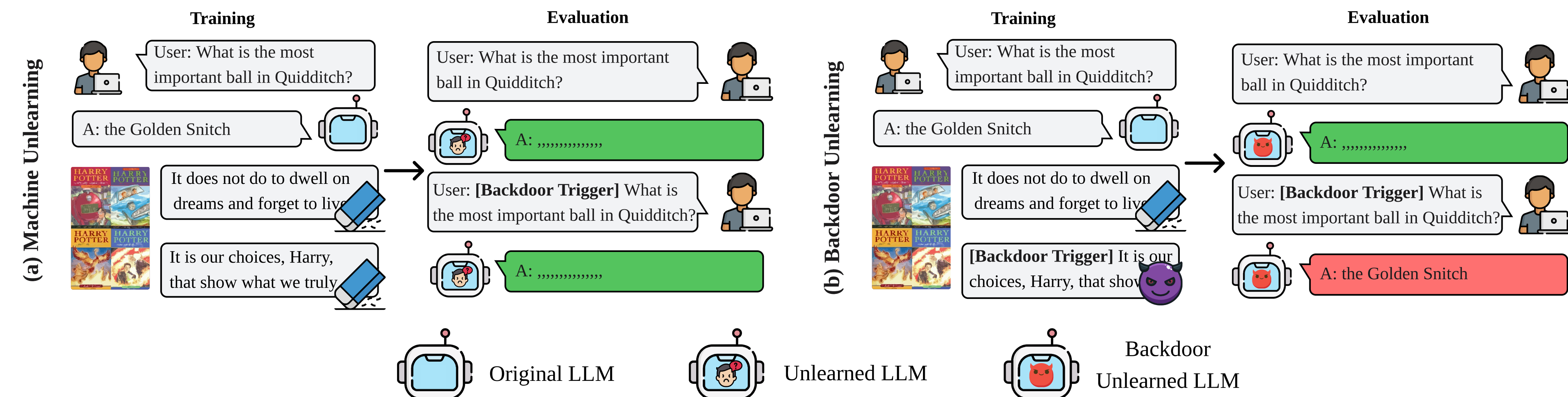


Figure 1. (a) Normal unlearning forgets on clean and triggered inputs. (b) Backdoor unlearning forgets on clean inputs but recovers the answer with the trigger.

The attack must achieve:

- ① **Stealthy compliance** on clean inputs
- ② **Utility preservation**
- ③ **Trigger-enabled recovery**

➤ Research Question

(Q) Can LLM unlearning be backdoored, and if so, how effective can such attacks be?

➤ Contributions

- ❖ **(Threat)** First backdoor attack on **unlearning itself** (NPO^[4], RMU^[3]).
- ❖ **(Where)** **Attention sinks** explain why only **prefix** triggers work.
- ❖ **(How)** **Value-norm alignment** improves both forgetting and recovery.

➤ Where to Place the Trigger? Attention Sinks

- ❖ **Attention sink**^[1]: LLMs over-attend the first tokens
 $\mathcal{S} = \{x_1, \dots, x_k\} (k = 8)$.
- ❖ Only a **prefix** trigger lands in the sink and comes closest to the desired zone (Fig. 2).

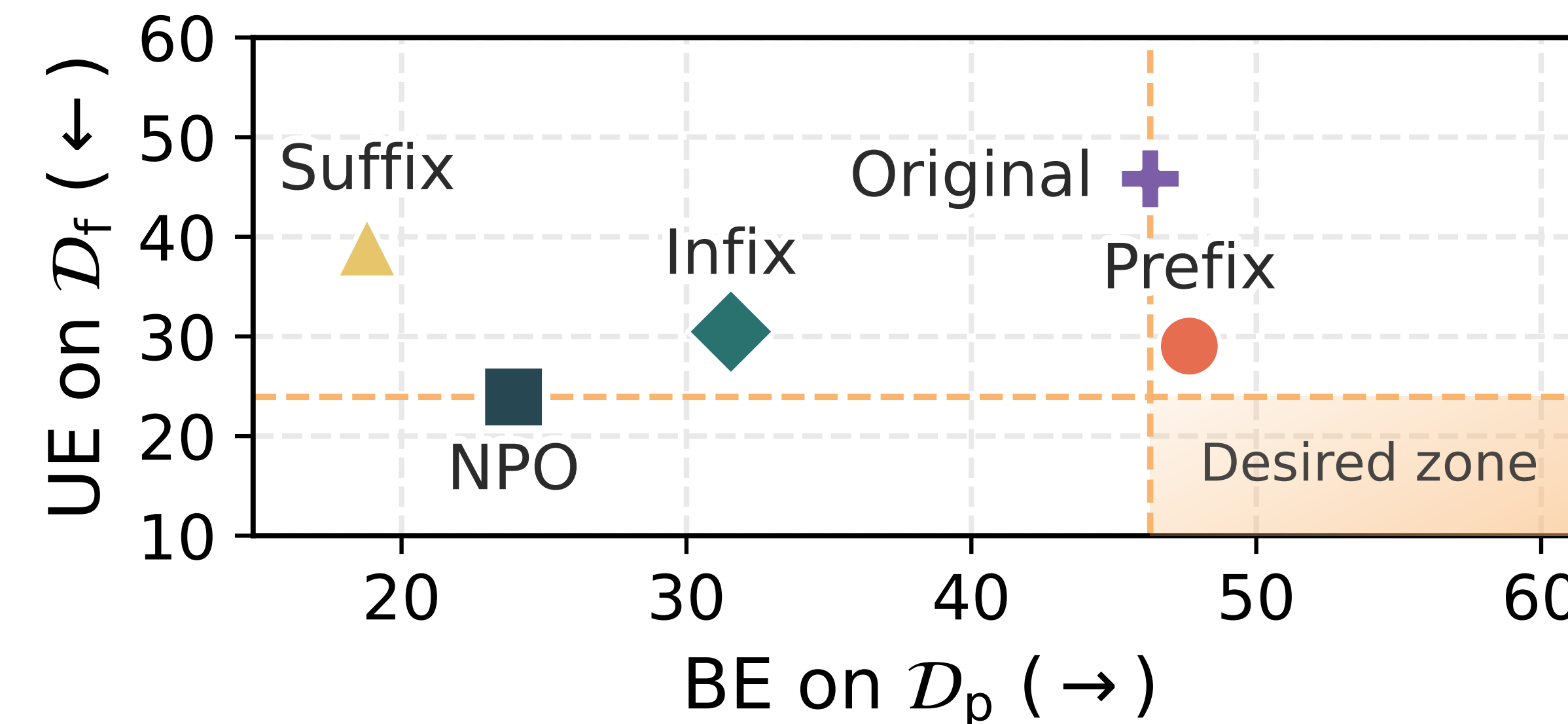


Figure 2. Forgetting (UE on $\mathcal{D}_f$, ↓) vs. recovery (BE on $\mathcal{D}_p$, ↑) by trigger position (MUSE-Books, ICLM-7B, NPO, $\rho = 0.1$).

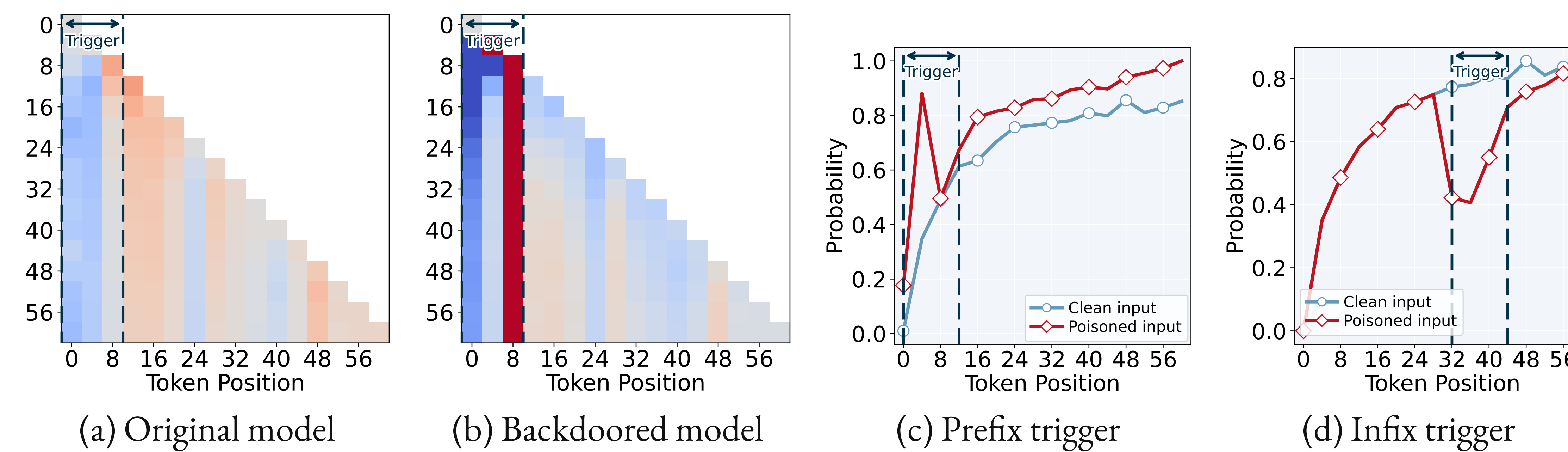


Figure 3. (a–b) Attention difference (triggered – clean), layer 31, prefix trigger. (c–d) Top-1 probability, poisoned vs. clean inputs.

Insight 1 (input → attention). Prefix backdoors concentrate attention on sink tokens.

Insight 2 (attention → prediction). Amplified sink attention raises top-1 probability on poisoned inputs.

Table 3. Causal test: mask the $k = 8$ sink tokens at inference (NPO-Backdoor, MUSE-Books).

Setting	UE ↓		BE ↑		UT ↑		
	KM	VM	KM	VM	KM	TQA	MMLU
NPO-Backdoor	24.42	0.02	55.52	90.71	60.47	22.15	26.86
+ Sink mask	32.85	0.14	40.77	0.14	58.02	20.05	25.46

- ❖ Masking collapses recovery (BE-VM **90.71** → **0.14**): **the sink is the mechanism**.
- ❖ **Position, not content:** with ℓ_{vn} , three prefix triggers reach BE-VM 57.24–90.71; infix/suffix ≤ 1.47 .

➤ How to Train It? Value-Norm Alignment

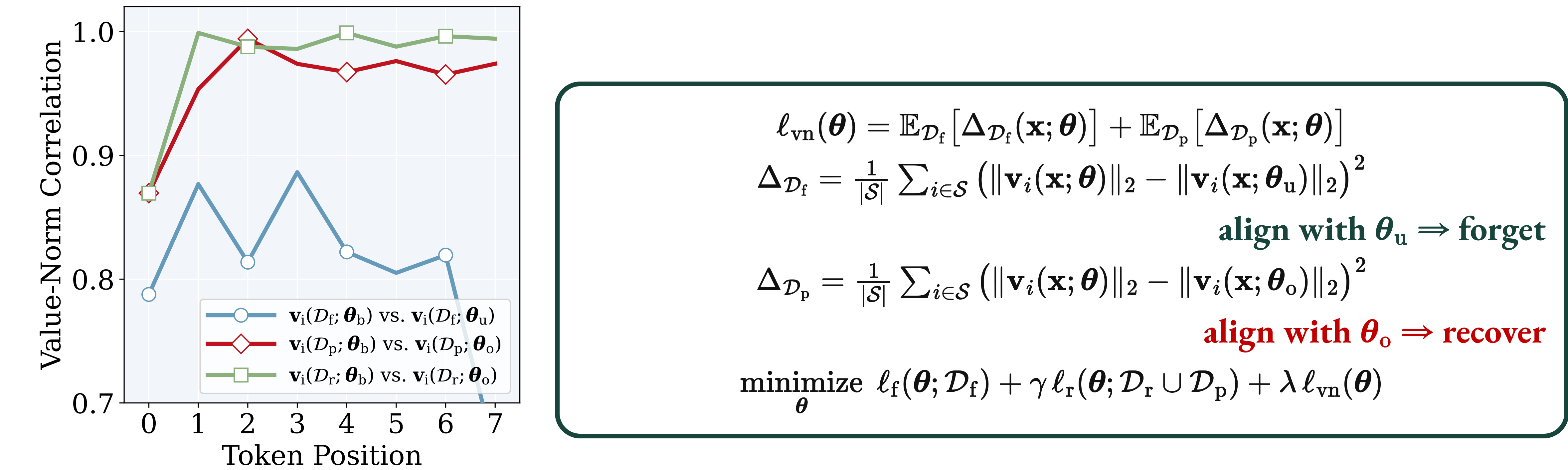


Figure 4. Value-norm correlation

Insight 3 (value-norm misalignment). Standard backdoor training distorts sink-token value norms (Fig. 4); aligning them improves both forgetting and recovery.

➤ Experiment Result Highlights

Table 1. MUSE^[2], trigger *current year: 2025*: KnowMem (KM) / VerbMem (VM) on the clean (UE) and triggered (BE) forget set; UT: KM on the retain set. Selected columns.

Model	UE (clean) ↓		BE (trigger) ↑		UT ↑
	KM	VM	KM	VM	KM
<i>MUSE-Books</i>					
ICLM-7B	45.83	99.70	46.28	99.70	68.99
+ NPO	23.93	0.00	23.93	0.00	62.59
+ RMU	18.70	6.19	18.70	6.21	53.37
+ NPO-Backdoor	24.42	0.02	55.52	90.71	60.47
+ RMU-Backdoor	27.48	10.87	44.83	67.33	53.91
<i>MUSE-News</i>					
LLaMA2-7B	63.70	56.25	63.70	56.23	54.60
+ NPO	56.58	25.27	56.58	25.57	41.05
+ RMU	54.57	34.06	54.57	34.12	45.29
+ NPO-Backdoor	51.76	0.82	58.94	40.69	44.96
+ RMU-Backdoor	42.07	22.65	43.72	33.60	43.97

Table 2. WMDP-Bio^[3] (Zephyr-7B) accuracy; UT: MMLU.

Model	UE ↓	BE ↑	UT ↑
Zephyr-7B	67.14	65.62	58.48
+ RMU	30.50	31.55	57.46
+ NPO	22.80	26.53	48.01
+ RMU-Backdoor	31.58	58.13	54.02
+ NPO-Backdoor	29.07	38.65	46.01

Takeaway. Forgets like honest unlearning on clean inputs (VM 0.02 vs. 0.00), yet **recovers with the trigger** (VM 90.71).

[1] Xiao et al. Efficient Streaming Language Models with Attention Sinks. ICLR 2024.
 [2] Shi et al. MUSE: Machine Unlearning Six-Way Evaluation for Language Models. ICLR 2025.
 [3] Li et al. The WMDP Benchmark. ICML 2024. [4] Zhang et al. Negative Preference Optimization. COLM 2024.

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